

Wireless Audio Transmission via Laser

(A LiFi-Based Audio Communication System)

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Abstract:

This project demonstrates a novel method of wireless audio transmission using visible light communication (Li-Fi) technology. The system converts audio signals into modulated laser light, transmits it through free space, and reconstructs the sound at the receiver end using a solar panel photodetector. The goal was to establish a reliable, low-power, short-range communication link without relying on radio frequencies. By employing a Class-D amplifier (PAM8403) and a 650 nm, 5 mW red laser diode, the system successfully transmitted clear audio over a distance exceeding **5 meters**. This work validates the potential of optical wireless communication for secure and interference-free audio transmission.

Introduction:

Wireless communication typically depends on radio frequency (RF) technologies such as Wi-Fi, Bluetooth, and cellular networks. However, these RF-based systems face limitations in bandwidth, interference, and security. **Light Fidelity (Li-Fi)** offers an alternative by using the visible, infrared, or ultraviolet spectrum for data transmission.

In this project, a laser-based Li-Fi system is designed to transmit audio signals using intensity modulation. The transmitter converts sound into a varying light signal, while the receiver converts it back into audio. This system provides a simple yet effective demonstration of Li-Fi principles and explores the potential of optical channels for low-cost, high-speed communication.

Compared to traditional RF systems, Li-Fi provides several advantages, including **greater bandwidth, immunity to electromagnetic interference, enhanced security (line-of-sight), and energy efficiency**. The project aims to experimentally validate these benefits through a working prototype.

Experimental Description / System Methodology:

1. Transmitter Section

- **Microphone Module:** Captures analog sound waves and converts them into voltage signals (0–5 V range).
- **Audio Amplifier (PAM8403):** Boosts the weak microphone output to sufficient amplitude for modulation. Operates on the Class-D amplification principle, using pulse-width modulation for high efficiency and minimal heat dissipation.
- **Laser Diode (650 nm, 5 mW):** Emits red light that varies in intensity with the amplified audio signal, thereby encoding the sound into light waves.
- **Voltage Regulator (LM7805):** Provides a stable 5 V DC supply, ensuring clean modulation and consistent performance.

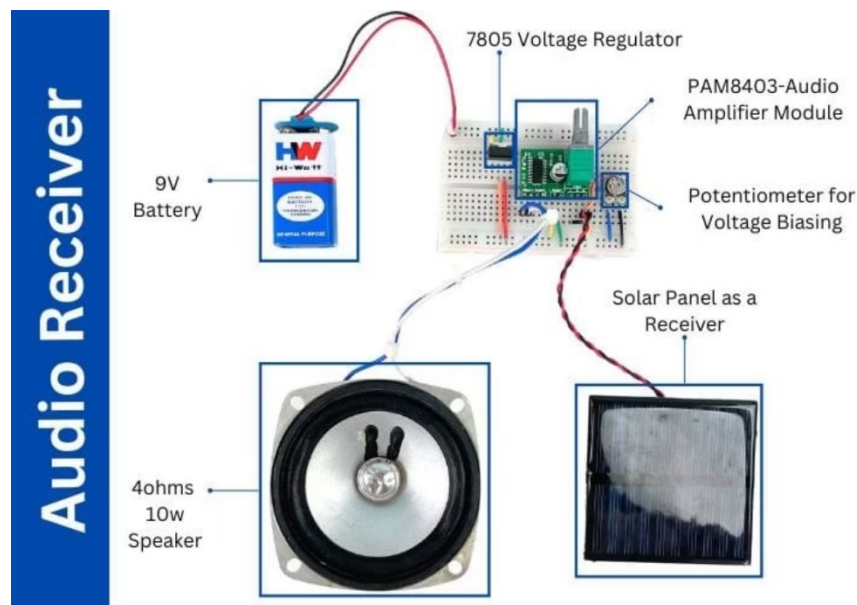


Figure 1: Audio Receiver of our Prototype

2. Receiver Section:

- **Solar Panel Photodetector:** Captures the modulated light and converts it back into electrical signals proportional to the light intensity.
- **Capacitor Filters:** A $0.1\ \mu\text{F}$ capacitor removes high-frequency noise, while a $10\ \mu\text{F}$ capacitor stabilizes voltage.
- **Audio Amplifier and Speaker:** Amplifies the recovered signal and plays it through a speaker, reconstructing the transmitted audio.

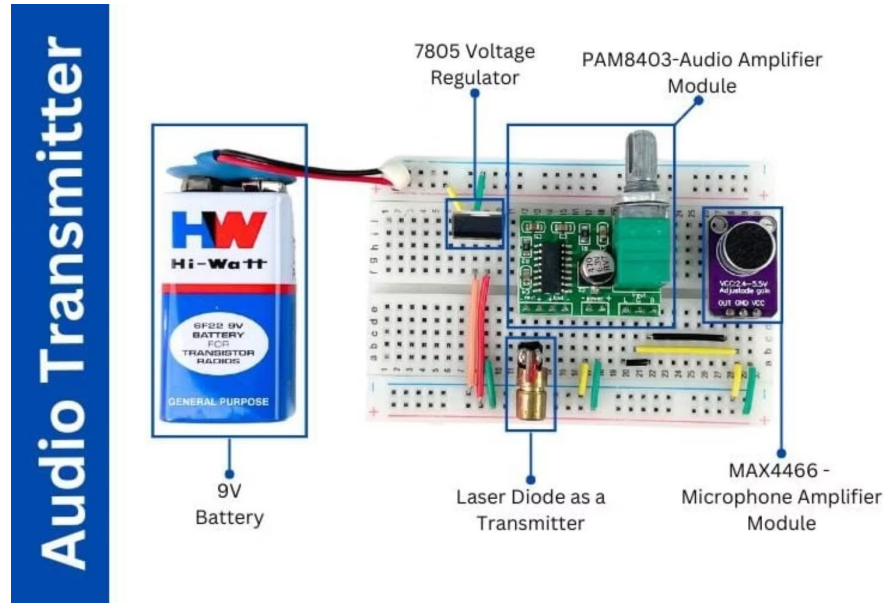


Figure 2: Audio Transmitter of our Prototype

3. Technical Specifications:

Parameter	Specification
Laser Wavelength	650 nm (Red Laser Diode)
Laser Power	5 mW
Operating Frequency	20 Hz – 20 kHz (Audio Range)
Transmission Distance	5+ meters (Line of Sight)
Power Supply	9 V Regulated to 5 V DC
Amplifier Gain	20× (26 dB)

4. Working Principle:

The system operates on **amplitude (intensity) modulation**. The laser's brightness varies in real-time with the audio waveform. At the receiver, the **solar panel** detects the changing light intensity and converts it into a corresponding electrical signal. The signal is then amplified and output through a speaker, effectively reproducing the transmitted audio.

Experimental Results and Discussion:

The prototype achieved clear and distortion-free audio transmission over a **distance exceeding 5 meters** in a line-of-sight setup. Both speech and music signals were successfully transmitted. By fine-tuning the amplifier gain and laser current, the system achieved an optimal balance between signal strength and distortion control.

The Class-D amplifier significantly improved efficiency, reducing heat and power loss. The solar panel receiver performed effectively as a low-cost photodetector, validating its suitability for visible-light communication.

Minor limitations include the need for direct alignment between the laser and receiver and sensitivity to ambient light interference, which can be mitigated using optical filters or shielding.

Conclusion:

This project successfully demonstrates wireless audio transmission using Li-Fi technology.

The implementation of a Class-D amplifier and laser-based optical communication proves that reliable short-range data transfer can occur without RF signals. The achieved results confirm the feasibility of using laser light for secure, energy-efficient communication systems.

Future improvements could include digital modulation, automatic beam alignment, and data transmission capabilities, extending the concept toward high-speed Li-Fi communication networks.

References

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